BOE-605	OPERATIONAL RESEARCH	L T P 3 0 0	3 Credits

Course Objectives:

CO1	To impart knowledge in concepts and tools of Operations Research.
CO2	To understand mathematical models used in Operations Research.
CO3	To apply these techniques constructively to make effective business decisions.
CO4	Develop mathematical skills to analyze and solve integer programming and network models arising from a wide range of applications.
CO5	Effectively communicate ideas, explain procedures and interpret results and solutions in written and electronic forms to different audiences.

MODULE I

Introduction:
Definition and scope of operations research (OR), OR model, solving the OR model, art of modelling, phases of OR study. Linear Programming:
Two variable Linear Programaming model and Graphical method of solution, Simplex method, Dual Simplex method, special cases of Linear Programming, duality, sensitivity analysis. Transportation Problems: Types of transportation problems, mathematical models , transportation algorithms, Assignment: Allocation and assignment problems and models, processing of job through machines.

MODULE II

Network Techniques: Shortest path model, minimum spanning Tree Problem, Max-Flow problem and Min-cost problem. Project Management:
Phases of project management, guidelines for network construction, CPM and PERT.

MODULE III

Theory of Games: Rectangular games, Minimax theorem, graphical solution of 2 x n or m x 2 games, game with mixed strategies, reduction to linear programming model.
Quality Systems: Elements of Queuing model, generalized poison queuing model, single server models. Inventory Control: Models of inventory, operation of inventory system, quantity discount. Replacement:
Replacement models: Equipments that deteriorate with time, equipments that fail with time.

Text Books:

1. Wayne L. Winston,"Operations Research" Thomson Learning,2003.
2. Hamdy H. Taha, "Operations Research-An Introduction" Pearson Education,2003.
3. R. Panneer Seevam, "Operations Research" PHI Learning, 2008.
4. V.K.Khanna, "Total Quality Management" New Age International, 2008.

Course Outcomes:

CO1	To be able to understand the application of OR and frame a LP Problem with solution
CO2	To be able to use different mathematical models and the solution procedures.
CO3	To understand the usage of game theory and Simulation for Solving Business Problems
CO4	To be able to design new simple models, like: CPM, MSPT to improve decision –making and develop critical thinking and objective analysis of decision problems
CO5	To be able to implement practical cases, by using TORA, WinQSB